1. docker compose version: Kiểm tra Docker Compose đã được cài chưa và phiên bản đang dùng

```
Microsoft Windows [Version 10.0.19045.6216]
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C:\Users\Student>docker compose version
Docker Compose version v2.22.0-desktop.2

C:\Users\Student>
```

2. docker compose config: Kiểm tra file cấu hình

```
Microsoft Windows [Version 10.0.19045.6216]
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T:\test>docker compose config
name: test
services:
  app:
    image: nginx
    networks:
      default: null
    ports:
      - mode: ingress
        target: 80
        published: "8080"
        protocol: tcp
networks:
  default:
    name: test_default

T:\test>
```

3. docker compose build: Build image từ Dockerfile

```
T:\test>docker compose build
[+] Building 0.0s (0/0)                                                                    docker:default

T:\test>
```

4. docker compose up: Chạy foreground (debug)

```
T:\test>docker compose up
[+] Running 8/8
 ⠿ app 7 layers [⠿⠿⠿⠿⠿⠿⠿]      0B/0B      Pulled                                                    8.2s
   ⠿ ec781dee3f47 Pull complete                                                                      2.7s
   ⠿ bb3d0aa29654 Pull complete                                                                      2.6s
   ⠿ 510ddf6557d6 Pull complete                                                                      1.2s
   ⠿ cde7a05ae428 Pull complete                                                                      2.4s
   ⠿ 587e3d84dbb5 Pull complete                                                                      3.6s
   ⠿ 3189680c601f Pull complete                                                                      3.8s
   ⠿ 5e815e07e569 Pull complete                                                                      3.9s
[+] Building 0.0s (0/0)                                                                    docker:default
[+] Running 2/2
 ⠿ Network test_default  Created                                                                     0.0s
 ⠿ Container test-app-1  Created                                                                     0.2s
Attaching to test-app-1
test-app-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
test-app-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
test-app-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
test-app-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
test-app-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
test-app-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
test-app-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
test-app-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
test-app-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: using the "epoll" event method
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: nginx/1.29.7
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: OS: Linux 5.10.102.1-microsoft-standard-WSL2
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
```

5. docker compose up –d: Chạy background (thường dùng)

```
T:\test>docker compose up -d
[+] Building 0.0s (0/0)                                                                    docker:default
[+] Running 2/2
 ✔ Network test_default  Created                                                                    0.0s
 ✔ Container test-app-1  Started                                                                    0.0s

T:\test>
```

6. docker compose up -d –build: Vừa build vừa chạy

```
T:\test>docker compose up -d --build
[+] Building 0.0s (0/0)                                                                    docker:default
[+] Running 1/0
 ✔ Container test-app-1  Running                                                                    0.0s

T:\test>
```

7. docker compose ps: Kiểm tra container đang chạy

```
T:\test>docker compose ps
NAME         IMAGE     COMMAND                                          SERVICE   CREATED         STATUS              PO
RTS
test-app-1   nginx     "/docker-entrypoint.sh nginx -g 'daemon off;'"   app       6 minutes ago   Up About a minute   0.
0.0.0:8080->80/tcp

T:\test>
```

8. docker compose logs –f: Xem log

```
T:\test>docker compose logs -f
test-app-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
test-app-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
test-app-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
test-app-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
test-app-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
test-app-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
test-app-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
test-app-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
test-app-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: using the "epoll" event method
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: nginx/1.29.7
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: OS: Linux 5.10.102.1-microsoft-standard-WSL2
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker processes
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 29
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 30
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 31
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 32
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 33
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 34
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 35
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 36
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 37
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 38
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 39
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 40
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 41
test-app-1  | 2026/04/02 12:31:35 [notice] 1#1: start worker process 42
```

9. docker compose exec <service_name> <command>: Vào container đang chạy

```
T:\test>docker compose exec app ls
bin   dev                  docker-entrypoint.sh  home  lib64  mnt  proc  run   srv  tmp  var
boot  docker-entrypoint.d  etc                   lib   media  opt  root  sbin  sys  usr

T:\test>
```

10. docker compose run <service_name> <command>: Chạy task một lần (migration, test…)

```
T:\test>docker compose run app ls
[+] Building 0.0s (0/0)                                                                                docker:default
[+] Building 0.0s (0/0)                                                                                docker:default
bin   docker-entrypoint.d   home   media  proc  sbin  tmp
boot  docker-entrypoint.sh  lib    mnt    root  srv   usr
dev   etc                   lib64  opt    run   sys   var

T:\test>
```

11. docker compose restart: Restart container

```
T:\test>docker compose restart
[+] Restarting 1/1
 ⠿ Container test-app-1  Started                                                                       0.8s

T:\test>
```

12. docker compose stop <service_name>: Stop container cụ thể

```
T:\test>docker compose stop app
[+] Stopping 1/1
 ⠿ Container test-app-1  Stopped                                                                       0.4s

T:\test>
```

13. docker compose rm <service_name>: Xóa container cụ thể

```
T:\test>docker compose rm app
? Going to remove test-app-1 Yes
[+] Removing 1/0
 ⠿ Container test-app-1  Removed                                                                       0.0s

T:\test>
```

14. docker compose down: docker compose down

```
T:\test>docker compose down
[+] Running 1/0
 ! Network test_default  Resource is still in use                                                      0.0s

T:\test>
```

15. docker compose down –v: Xóa luôn volume

```
T:\test>docker compose down -v
[+] Running 1/0
 ! Network test_default  Resource is still in use                                                      0.0s

T:\test>
```